

Result

Size

Time

Cycles

GPU

SM Frequency

Process

Attributes

Current

1185 - matmul_k_tiled

(1, 3125, 1)x(16, 16, 1)

580.64 us

1,521,886

0 - NVIDIA GeForce RTX 4070 Ti

2.62 Ghz

[72087] python3.14



Summary

Details

Source

Context

Comments

Raw

Session

Compare

Tools

View

Export



This table shows all results in the report. Use the column headers to sort the results in this report. Double-click a result to see detailed metrics. Double-click on demangled names to rename it.

ID	Estimated Speedup [%]	Function Name	Demangled Name	Duration (6,402.88 us)	Runtime Improvement [us] (3,383.40 us)	Compute Throughput	Memory Throughput [MB/s]	# Registers [register/thread]	Grid Size
0	52.96	matmul_k_tiled	matmul_k_tiled(floa...	580.64	307.50	64.59	64.59	38	1, 31
1	53.31	matmul_k_tiled	matmul_k_tiled(floa...	577.12	308.12	65.01	65.01	38	1, 31
2	52.61	matmul_k_tiled	matmul_k_tiled(floa...	584.12	307.38	64.16	64.16	38	1, 31
3	53.26	matmul_k_tiled	matmul_k_tiled(floa...	577.12	307.39	64.95	64.95	38	1, 31
4	53.45	matmul_k_tiled	matmul_k_tiled(floa...	575.12	307.50	65.18	65.18	38	1, 31
5	53.15	matmul_k_tiled	matmul_k_tiled(floa...	579.12	308.09	64.82	64.82	38	1, 31
6	52.71	matmul_k_tiled	matmul_k_tiled(floa...	583.12	307.49	64.28	64.28	38	1, 31
7	52.85	matmul_k_tiled	matmul_k_tiled(floa...	581.12	307.49	64.46	64.46	38	1, 31
8	53.12	matmul_k_tiled	matmul_k_tiled(floa...	578.12	307.51	64.78	64.78	38	1, 31
9	50.83	matmul_k_tiled	matmul_k_tiled(floa...	604.12	307.44	61.99	61.99	38	1, 31
10	53.11	matmul_k_tiled	matmul_k_tiled(floa...	578.12	307.50	64.77	64.77	38	1, 31

The following performance optimization opportunities were discovered for this result. Follow the rule links to see more context on the Details page.

Note: Speedup estimates provide upper bounds for the optimization potential of a kernel assuming its overall algorithmic structure is kept unchanged.

[L1TEX Global Load Access Pattern](#)

Est. Speedup: 52.96%

The memory access pattern for global loads from L1TEX might not be optimal. On average, only 5.8 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the [Source Counters](#) section for uncoalesced global loads.

► Key Performance Indicators

[L1TEX Global Load Access Pattern](#)

On average, each warp of this workload spends 27.6 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local/global/surface texture) operation. Find the instruction